Cpts 317. 12/5

Final Exam Review.

4 pages of single-sided cheating sheets are allowed.

Final Exam covers contents after midterm.

{ don't make mistakes on simple problems
{ try your best on hard ones.

---

go through your hw problems.

Basics —— c.f. grammar, PDA, TMs.

B1. Given a grammar, describe the language generated by the grammar.

Ex: $S \longrightarrow 0S1 \mid 01$

the language $= \{0^n 1^n : n \geq 1\}$

Ex: $S \longrightarrow AB$

$A \longrightarrow 0A \mid 00$

$B \longrightarrow B1 \mid 1$

the language $= \{0^n 1^m : n \geq 2, m \geq 1\}$

Ex: $S \longrightarrow 0AS\,A0 \mid \lambda$

$A \longrightarrow 1A \mid 1$

describe all strings generated by the grammar.

$$\underbrace{0A\;0A}_{0A\text{-blocks}} \cdots \underbrace{A0A0}_{A0\text{-blocks}}$$

pattern:

$$\underbrace{0\,1^+\,0\,1^+ \cdots\cdots 1^+\,0\,1^+\,0}_{\text{Even \# of 0's.}}$$

B3. Elimination of Λ-productions,

Elimination of unit-productions.

(look at class notes & hw).

B3'. Chomsky Normal Form. (CNF).

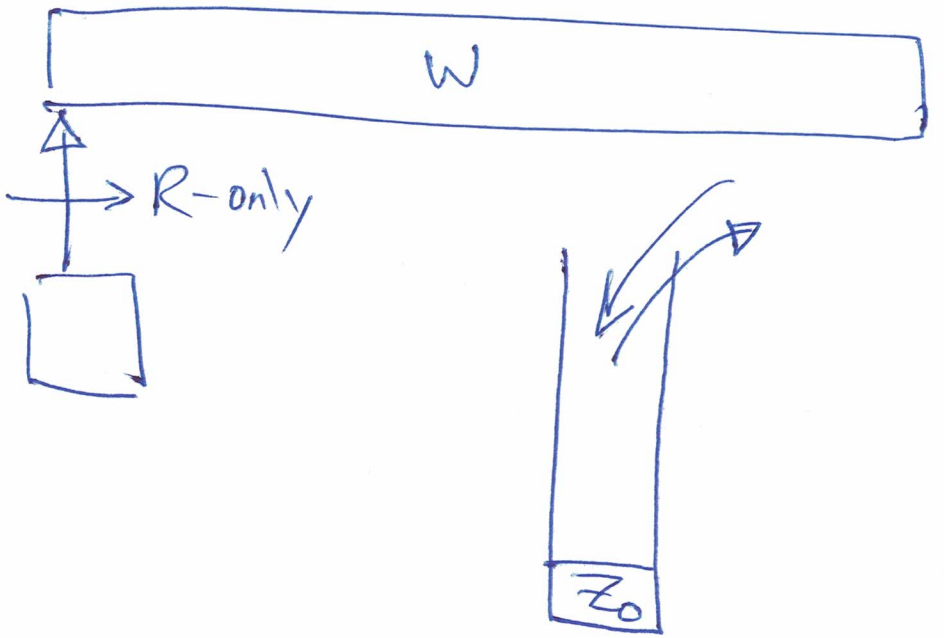
Ex. $S \to aSb \mid ab$. convert it to CNF.

$A \to a$ $\quad$ $S \to AC$

$B \to b$ $\quad$ $C \to SB$

$\cancel{S \to ASB}$

B4. PDA's.

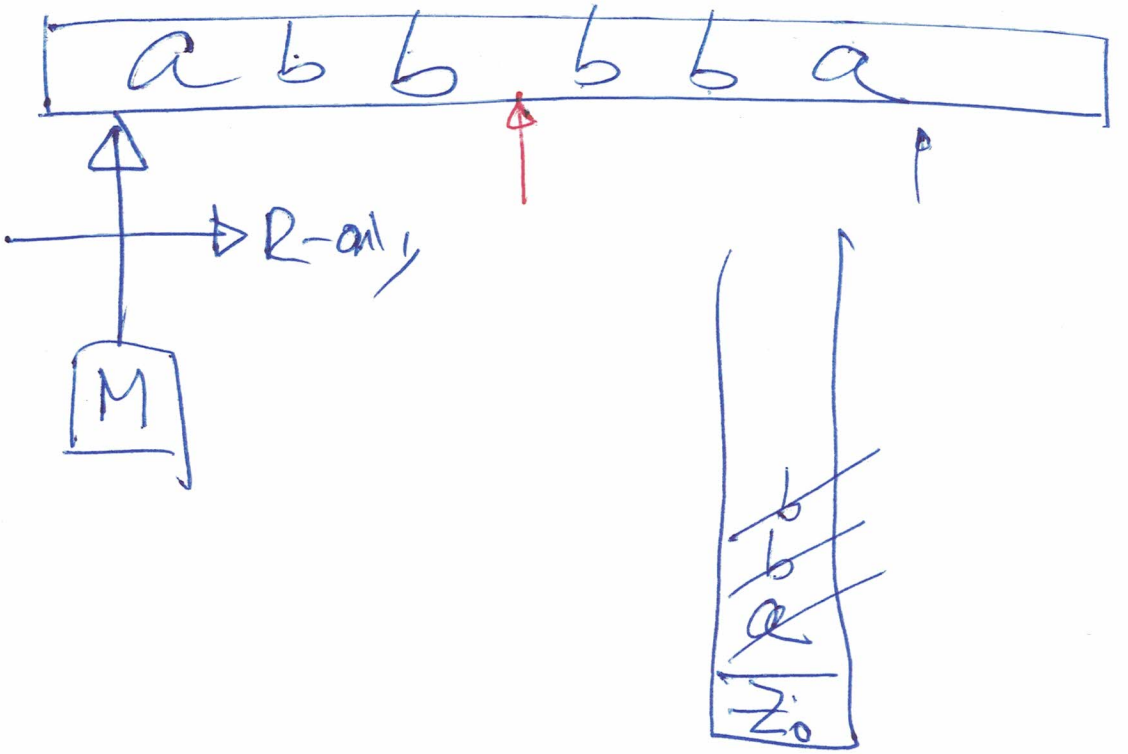


From a language, describe in English a PDA to accept the language.

// PDA can "guess".
//
//

Ex. Describe a PDA to accept

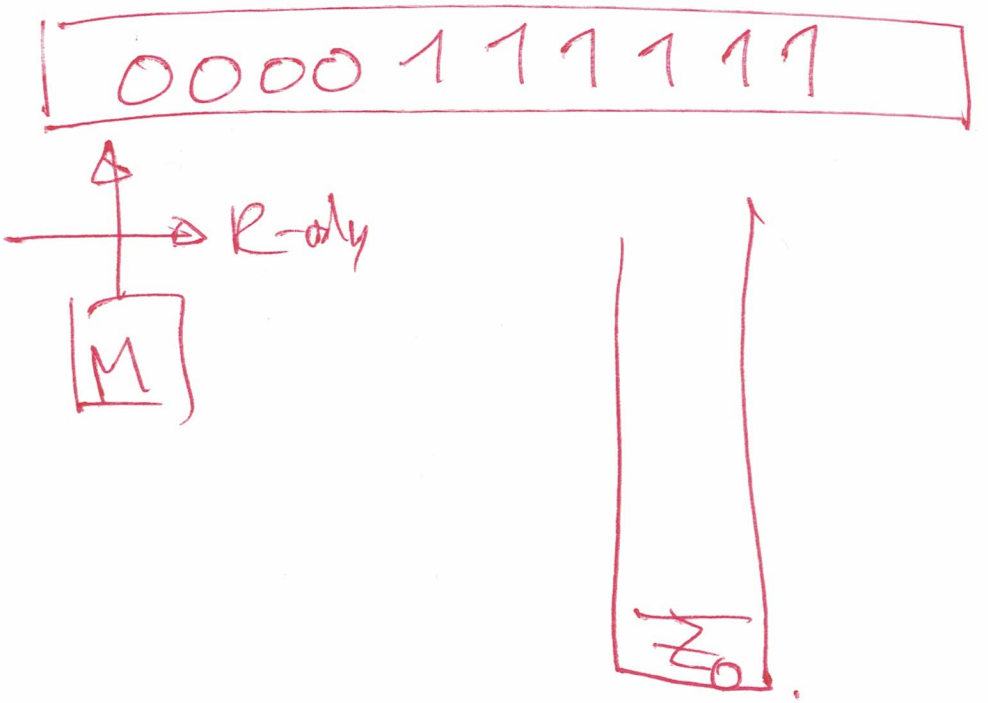
$$\{ x x^R : x \in \{a,b\}^* \}$$



Solution: On an input word, M works as follows. It pushes every symbol it reads onto the stack. Some moments later, it guess the head is in the middle of the input. Then it switches the mode: for every symbol it reads, it pops the stack and compare with the symbol read (making sure that they are the same). At the end of input, if the stack is empty M says yes on the input. Clearly M is a PDA accepting the language. $\boxtimes$

Ex. Describe in English a PDA to accept

$$\{0^{2n}1^{3n} : n \geq 0\}$$



Key: read two 0's at a time, & push three 1's at the same time!

Write it in English!

B5. Explicit construction of PDA's.
(use δ - instructions). look at hw.

B6. Conversion from grammar to PDA.
(look at hw).

~~(B7) —— Conversion from PDA to grammar~~

B7. using PL to prove a language is NOT c.f.
(100% prob. you have the problem on Exam).
(you shall write rigorously.)

tonight:

Show that $L = \{a^n b^n c^n : n \geq 0\}$ is not c.f.

⋮

B8. Write TM code to recog. a language:

Ex: $\mathcal{L} = (aa)^* b$. Can you write TM code:

$\boxed{aaaabBB\ldots}$

$$\delta(q_0, a) = (q_1, a, R)$$
$$\delta(q_1, a) = (q_0, a, R)$$
$$\delta(q_0, b) = (q_2, b, R)$$
$$\delta(q_2, B) = (q_{happy}, B, S).$$

q_{happy} is accepting.

Ex: (tonight). $\mathcal{L} = (aaa)^*(bb)^*c$. Write TM code.